INCH-POUND

A-A-59274
18 June 2003

COMMERCIAL ITEM DESCRIPTION

GENERATOR, SIGNAL
9kHz – 3200 MHz

The General Services Administration has authorized the use of this Commercial Item Description, for all federal agencies.

1. SCOPE. This Commercial Item Description (CID), describes a Signal Generator in which only those manufacturers meeting or surpassing the following requirements are supplied per this CID. Instruments meeting this CID must be capable of operation within the equipment environment specified herein.

2. CLASSIFICATION

2.1 Class. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following salient characteristics. The equipment shall be Class 4 in accordance with MIL-PRF-28800.

3. SALIENT CHARACTERISTICS

3.1 Bid Sample Testing. The equipment shall meet all the requirements of this CID and shall be bid sample tested, if so requested, within the accuracies, limits and specifications herein.

3.2 Safety and environmental. The equipment shall meet all safety and environmental requirements as specified the latest version MIL-PRF-28800 for the classification as stated herein.

3.3 Order of Precedence: In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications), specification sheets or MS standards the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Automatic Test & Measurement Branch WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611.

AMSC N/A FSC 6625

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3.4 Mean Time Between Failures (MTBF): The equipment Mean Time Between Failures (MTBF) shall be a minimum of 4,000 hours. This is based upon an anticipated usage of 2,000 hours per year.

3.5 Calibration and Maintenance Adjustments: The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

3.6 Frequency Related Specifications

3.6.1 Frequency Range/Resolution. The frequency range and resolution of this signal generator shall be as follows: Range: 9kHz – 3200MHz and Resolution: 0.001Hz

3.6.2 Accuracy (within 1 year of adjustment). The accuracy of this instrument shall be as follows: $\pm$ 0.15 X 10^{-6} X carrier frequency (Hz)

3.6.3 Switching Speed. The switching speed of this instrument shall be as follows:

<1001 MHz: < 75 ms
=1001 MHz: < 100 ms

3.7 Internal Reference Oscillator.

3.7.1 Accuracy and Stability. The accuracy and stability shall be as follows:

Aging: < $\pm$ 0.1 ppm/yr
Temperature: < $\pm$ 0.01 ppm
Line voltage: < $\pm$ 0.1 ppm

3.7.2 Output. The output shall be 10 MHz, > 0.5 V^{rms} level into 50 Ohms.

3.8 Output Related Specifications.

3.8.1 Output Range. The output range on this instrument shall be +10 dBm to -136 dBm.

3.8.2 Display Resolution. The display resolution on this instrument shall be 0.1 dB.

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3.8.3 Standing Wave Ratio (SWR). The SWR on this instrument with output <-6 dBm shall be as follows:

< 249 kHz : < 2.5:1
< 2500 MHz : < 1.5:1
= 3200 MHz : < 2.0:1

3.8.4 Output Impedance. The output impedance shall be 50 Ohms.

3.8.5 Reverse Power Protection. The reverse power protection shall be as follows:

=2000 MHz : 50 watts
=3200 MHz : 25 watts

3.8.6 Accuracy. The instrument shall have an output accuracy as follows:

=2500 MHz : ± 1.0 dB
=3200 MHz : ± 1.5 dB

3.9 Spectral Purity.

3.9.1 Harmonics. The instrument shall have <-30 dBc harmonics with output = +4 dBm.

3.9.2 Subharmonics. The instrument shall have subharmonics as follows with output = +4 dBm:

< 1001 MHz : <-60 dBc
= 3200 MHz : <-50 dBc

3.9.3 Nonharmonics. The instrument shall have nonharmonic levels as follows at = 5 kHz and output = +4 dBm:

< 249 MHz : <-55 dBc
< 1001 MHz : <-60 dBc
< 2001 MHz : <-54 dBc

3.9.4 Residual FM. The instruments shall have residual FM as follows:

< 249 MHz : < 7 Hz
< 501 MHz : < 4 Hz
< 1001 MHz : < 7 Hz
< 2001 MHz : < 14 Hz

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3.9.5 Single Side Band (SSB) Phase noise. The instrument shall have SSB phase noise at 20 kHz offset as follows:

500 MHz : <- 120 dBc/ Hz
1000 MHz : <- 116 dBc/ Hz
2000 MHz : <- 110dBc/ Hz
3000 MHz : <- 106 dBc/ Hz

3.10 Amplitude Modulation.

3.10.1 Range. The amplitude modulation range shall be 0 to 100% (output = +4 dBm).

3.10.2 Resolution. The amplitude modulation resolution shall be 0.1%.

3.10.2 Accuracy. The amplitude modulation accuracy at 1 kHz shall be ± 5% of setting ± 1.5%.

3.10.3. Rates. The amplitude modulation rates shall be as follows:

Internal: 400 Hz or 1 kHz
External: DC: dc to 25 kHz (typical, 3 dB BW)
AC: 1 Hz to 25 kHz (typical, 3 dB BW)

3.10.4 Distortion. The amplitude modulation distortion shall be as follows: (1kHz rate, THD + N, 0.3 to 3 kHz BW)

30% AM : < 2%
70% AM : <3%

3.11 Frequency Modulation.

3.11.1 Frequency Modulation Peak Deviation. The frequency modulation peak deviation shall be as follows: (rates > 25 HZ ac FM)

<249 MHz : 0 to 200 kHz
< 501 MHz : 0 to 100 kHz
< 1001 MHz : 0 to 200 kHz
< 2001 MHz : 0 to 400 kHz
< 3200 MHz : 0 to 800 kHz

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3.11.2 Frequency Modulation Resolution. The frequency modulation resolution shall be as follows:

For = 10% peak deviation
< 2001 MHz : 10 Hz
= 2001 MHz : 20 Hz

For > 10% peak deviaton
< 2001 MHz : 100 Hz
= 2001 MHz : 200 Hz

3.11.3 Frequency Modulation Deviation Accuracy. The frequency modulation deviation accuracy shall be as follows: (internal 1 kHz rate)

< 1001 MHz : ±3% of FM deviation ±30 Hz
< 2001 MHz : ±3% of FM deviation ±60 Hz
= 3200 MHz : ±3% of FM deviation ±120 Hz

3.11.4 Frequency Modulation Rates. The frequency modulation rates shall be as follows:

Internal: 400 Hz or 1 kHz
External: DC: dc to 150 kHz (typical, 3 dB BW)
AC: 1 Hz to 150 kHz (typical, 3 dB BW)

3.11.5 Frequency Modulation Distortion. The frequency modulation distortion shall be as follows: (1 kHz rate, THD + N, 0.3 to 3 kHz BW)

< 1001 MHz : < 1% at deviations > 4 kHz
< 2001 MHz : < 1% at deviations > 8 kHz
= 3200 MHz : < 1% at deviations > 16 kHz

3.11.6 Carrier Frequency Accuracy. The frequency modulation carrier frequency accuracy shall be as follows:

< 1001 MHz : ± 100 (typical 40) Hz; deviations < 10 kHz
< 2001 MHz : ± 200 (typical 80) Hz; deviations < 20 kHz
= 3200 MHz : ± 400 (typical 160) Hz; deviations < 40 kHz

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3.12 Phase Modulation.

3.12.1 Phase Modulation Peak Deviation. The phase modulation peak deviation shall be as follows:

<249 MHz : 0 to 10 radians
< 501 MHz : 0 to 5 radians
< 1001 MHz : 0 to 10 radians
< 2001 MHz : 0 to 20 radians
= 3200 MHz : 0 to 40 radians

3.12.2 Phase Modulation Resolution. The phase modulation resolution shall be as follows:

< 2001 MHz : 0.01 radians
= 2001 MHz : 0.02 radians

3.12.3 Phase Modulation Deviation Accuracy. The phase modulation deviation accuracy shall be as follows: (internal 1 kHz rate)

< 1001 MHz : ±3% of deviation ±0.05 radians
< 2001 MHz : ±3% of deviation ±0.1 radians
= 3200 MHz : ±3% of deviation ±0.2 radians

3.12.4 Phase Modulation Rates. The phase modulation rates shall be as follows:

Internal: 400 Hz or 1 kHz
External: 20 Hz to 10 kHz (typical, 3 dB BW)

3.12.5 Phase Modulation Distortion. The phase modulation distortion shall be as follows: (1 kHz rate)

< 1001 MHz : < 1% at deviations = 3 radians
< 2001 MHz : < 1% at deviations = 6 radians
= 3200 MHz : < 1% at deviations = 12 radians

3.13 Modulation Source Specifications.

3.13.1 Internal Modulation Source. The internal modulation source shall be 400 Hz, front panel BNC connector provided at nominally 1 Vpk into 600 Ohms.

3.13.2 External Modulation Source. The external modulation source shall be 1 Vpk into 600 Ohms (nominal) required for full scale modulation.

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3.14 Pulse Modulation Related Specifications.

3.14.1 Pulse Modulation On/Off Ratio. The pulse modulation On/Off ratio shall be as follows:

< 2000 MHz : > 80 dB
= 3200 MHz : > 70 dB

3.14.2 Pulse Modulation Rise/Fall Times. The pulse modulation rise/fall time shall be < 10 ns.

3.14.3 Pulse Modulation Repetition Rate. The maximum pulse modulation repetition rate shall be 10 MHz.

3.14.4 Pulse Modulation Delay. The pulse modulation delay shall be less than 60 ns.

3.14.5 Pulse Modulation Input.The pulse modulation input shall be TTL level (± 15V max).

3.15 Compatibility Related Specifications.

3.15.1 Remote Operation. Equipment shall interface with GPIB (IEEE-488.2) Listen and Talk.

3.15.2 Storage Registers. Equipment shall have 300 storage registers with sequence and register number displayed.

3.16 Environmental Related Specifications.

3.16.1 Operating Temperature. The operating temperature shall be 0°C to 40°C.

3.16.2 Non-Operating Temperature. The non-operating temperature shall be -40°C to + 71°C.

3.16.3 Operating Humidity. The operating humidity shall be 95% ±5% RH at + 10°C to +30°C. The operating humidity shall be 75% ±5% RH at +30°C to +40°C.

3.17 Physical Dimensions and Characteristics.

3.17.1 Dimensions and Weight. The instrument shall have the envelop dimensions of 6.5" H (16.5 cm) x 13" W (33 cm) x 14.6"L (36.8 cm) and a weight not to exceed 15 lb. (7 kg).

4. REGULATORY REQUIREMENTS

4.1. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). These are consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

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5. PRODUCT CONFORMANCE

5.1 Product Conformance. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Pre-Conditioning Requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase

6. PACKAGING. Preservation, packing, labeling, and marking shall be as specified in the contract or order. The Equipment shall have an identification plate permanently affixed in a readable place, which includes the information contained in MIL-PRF-28800, Section 3.

7. NOTES.

7.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in conversion tables contained in Federal Standard No. 376 and all other requirements of this CID are met. If the manufacturer elects to use metric dimensions as the basis upon which to derive or base a mechanical feature, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

7.2 Source of Documents.

7.2.1 Government Documents. Military Specifications, Standards and Handbooks referenced are available from the DODSSP, 700 Robbins Ave, Bldg. 4D, Philadelphia, PA 19111-5094. Phone 215-697-2179.

7.2.2 The Code of Federal Regulations (CFR). CFR documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.2.3. IEEE standards. IEEE standards may be obtained by contacting IEEE Operations Center, 445 Hoes Lane, PO Box 1331, Piscataway, NY, 08854-1331. Phone 1-800-678-4333 (Toll-Free, US & Canada), 1-732-981-0060, Fax: +1 732 981 9667 www.ieee.org.

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MILITARY INTERESTS:

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Air Force-99

Preparing Activity
Air Force-84

Agent:
Air Force-99

Project No. 6625-0924

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